

Read me first — sim re-entry (540, July 2026)

Last synced: 2026-07-30

OPORD: `../3-Master_Plan/re_entry/model_OPORD.md`

Your checklist: `../3-Master_Plan/re_entry/CHARLES_CHECKLIST.md`

Before this file's knockout table: read `../3-Master_Plan/re_entry/04_Pass_A_and_Pass_B_in_Plain_English.md` — full sentences on Pass A/B, empirical vs simulated, assignment, and which scripts to run. This README is the **contract / shorthand** after that.

If you feel lost: read `../3-Master_Plan/re_entry/00_READ_ME_FIRST.md` (docs 01–04) before opening sim code.

What to open vs what to ignore

Use daily (re-entry)	Ignore for daily work
This file + <code>540_three_step_sim.ipynb</code>	<code>tier1_cell10_playground_run.py</code> (LEGACY 538 widget UI)
<code>scripts/hero_model_reset_bundle.py</code> (Pass A)	Archived notebooks under <code>sports/archive/</code>
<code>scripts/540_rho_ablation_bundle.py</code> (Pass B)	<code>539_alex_model.ipynb</code> (older Alex lab; not 540 path)
<code>tier1_pool_assignment.py</code> , <code>tier1_generative_eda.py</code> , <code>tier1_sim_config.py</code>	Extending CELL 10 widgets

Engines (`tier1_*`) are **shared libraries**. Bundles are **thin runners**. Playground = old UI around the same engines.

Revolution vs evolution

| Do | Don't | |--|---| |

Import

`tier1_pool_assignment.py`,
`tier1_generative_eda.py`,
`tier1_sim_config.py` |

Extend archived notebooks

(`538D`, `538`, `537`, `535`) |

| Thin scripts / `540_*`

notebook | Copy-paste 538D

cells | | Answer Alex-sized

questions (knockouts,

ablations) | Rivanna sweeps,

bin-for-bin hero match as v1

gate |

Archived notebooks in

`sports/archive/` are

SCOUT-era lab reference —

read only, do not edit.

Charles's three-step pipeline

1. **Assign** — build rosters (ρ or sort-and-chop)
2. **Score** — $S_i = A_i - \lambda \cdot L_C$ (Alex v1: $(B-D)=-L_C$ in the score; λ lives here)
3. **Select** — winner rule: top K by S_i (later: soft / stochastic draw)

Knockouts (shorthand — only after doc 04)

Full sentences: `../3-Master_Plan/re_entry/04_Pass_A_and_Pass_B_in_Plain_English.md`.

Pass	What toggles	Status
A	$\lambda=0$ vs congestion in S_i (winner rule fixed)	Re-run → <code>HEROs_and_PASSes/PASS_A_*</code>
B	ρ low / ρ high / sort-and-chop; fixed score + select	Re-run → <code>HEROs_and_PASSes/PASS_B_*</code>

Three-step assignment (step 1 only)

0. Draw latent abilities A_i
1. Draw team targets T_j (e.g. Uniform on $[-0.5, 0.5]$)
2. Sequential soft match with roster caps:

$$\pi_{ij} \propto \exp\left(-\rho \cdot \frac{(A_i - T_j)^2}{2\sigma^2}\right) \times (n_j + k)^\alpha$$

Optional preferential attachment: `USE_PREFERENTIAL_ATTACHMENT = True` → $\alpha > 0$; default **False**.

ρ	Meaning
ρ = 0	Uniform among teams with roster room (max mixing)
ρ = 1	Moderate assortativity (≡ legacy $\tau = \sigma \approx 0.65$)
ρ high	Sharper match to nearest T_j
sort-and-chop	Separate code path — 537-style global sort + equal slices; max assortativity benchmark , not $\rho \rightarrow \infty$

σ (`ASSIGNMENT_SIGMA` , default **0.65**) is a fixed scale — not the user-facing knob.
Legacy docs use τ (**temperature**) with **opposite** intuition (small τ = assortative). **540+ uses ρ only** in prose and exports.

Hero estimand lock

Field	Value
Outcome	<code>Y_draft</code> , mean draft rate per bin
X (hero)	<code>poolq_loo</code>
Bins	16 quantile
Winsor	(0.01, 0.99)
min_minutes	20
Seasons	2011–2021

Slug: `empirical_ppm_poolq_loo_16quantile_winsor0199_min20_2011`

Active files

Role	Path
Empirical	<code>530_sports_pipeline.ipynb</code> , <code>sports_pipeline/</code>
Sim engines	<code>tier1_pool_assignment.py</code> , <code>tier1_generative_eda.py</code> , <code>tier1_sim_config.py</code>
Pass A bundle	<code>scripts/hero_model_reset_bundle.py</code> → <code>re_entry/HEROs_and_PASSes/PASS_A_*</code>
Pass B bundle	<code>scripts/540_rho_ablation_bundle.py</code> → <code>re_entry/HEROs_and_PASSes/PASS_B_*</code>
Notebook	<code>540_three_step_sim.ipynb</code>
Binding	<code>3-Master_Plan/BINDING_Selection_is_its_own_step.md</code>
Slide	<code>3-Master_Plan/re_entry/Model.pdf</code>

Export naming

```
3-Master_Plan/re_entry/HEROs_and_PASSes/  
HERO_*           # empirical hero PNG(s)  
PASS_A_*         # λ knockout (PNG, CSVs, summary)  
PASS_B_*         # ρ ablation (PNG, CSVs, summary, README)
```

Pass B artifacts: `PASS_B_generative_rho_{low,moderate,high,very_high}_16quantile.csv` , `PASS_B_generative_sort_chop_16quantile.csv` , `PASS_B_rho_ablation_summary.txt` , `PASS_B_rho_ablation_caption.txt` , `PASS_B_README.txt` , `PASS_B_*.png` .

Anti-patterns

- Opening `archive/` notebooks as the daily workspace
- Merging **L_net** and **S_i** in comments or claims
- Saying ρ ablation **proves** hero bin-for-bin match
- Using τ in new user-facing text (archive reference only)

When stuck

Question	Read
What are we proving?	<code>re_entry/02</code> , <code>Model.pdf</code>
Assignment math (legacy τ)	<code>documents/Tier1_Presorting_Design_Note.md</code>
Widget-era manual	<code>documents/538_Cell10_Generative_Manual.md</code> (archive context)
Implementation	<code>tier1_pool_assignment.py</code> — <code>soft_assign</code> , <code>simulate_generative_rosters</code>